



ALTA DISPONIBILITAT DE DHCP AMB KEA

NF1.1_PR203



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2n ASIX

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Part 1: Preparació dels Dos Servidors

Servidor primari amb hostname i la IP 172.24.63.10 (num. Llista 63)

```
smartinez6@kea-server-shail: x smartinez6@kea-server2-shail: x + v
smartinez6@kea-server-shailmartinez:~$ hostnamectl
Static hostname: kea-server-shailmartinez
Icon name: computer-vm
Chassis: vm
Machine ID: a1c33290b1c24a8b8132f2bb489e5737
Boot ID: da29125850374e4388303fb3447d2bfff
Virtualization: oracle
Operating System: Ubuntu 26.04.1 LTS
Kernel: Linux 7.0.0-34-generic
Architecture: x86_64
Hardware Vendor: innotek GmbH
Hardware Model: VirtualBox
Hardware Version: 1.2
Firmware Version: VirtualBox
Firmware Date: Fri 2006-12-01
Firmware Age: 19y 10month 4d
smartinez6@kea-server-shailmartinez:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:d4:73:fa brd ff:ff:ff:ff:ff:ff
    altname enx080027d473fa
    inet 172.24.63.10/24 brd 172.24.63.255 scope global enp0s3
        valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fed4:73fa/64 scope link link_proto kernel_ll
        valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:04:62:db brd ff:ff:ff:ff:ff:ff
    altname enx0800270462db
```

Servei de kea-dhcp4 i kea-ctrl-agent parat.

```
smartinez6@kea-server-shail: x smartinez6@kea-server2-shail: x + v
smartinez6@kea-server-shailmartinez:~$ sudo systemctl stop kea-dhcp4-server
smartinez6@kea-server-shailmartinez:~$ sudo systemctl stop kea-ctrl-agent
smartinez6@kea-server-shailmartinez:~$ sudo systemctl status kea-dhcp4-server
o kea-dhcp4-server.service - Kea IPv4 DHCP daemon
   Loaded: loaded (/usr/lib/systemd/system/kea-dhcp4-server.service; enabled; preset: enabled)
   Active: inactive (dead) since Mon 2026-10-05 21:25:14 UTC; 1min 53s ago
   Duration: 25min 4.511s
   Invocation: db3d60c73d99d45fbc8c6f4ce4f5f5fd
   Docs: man:kea-dhcp4(8)
   Process: 870 ExecStart=/usr/sbin/kea-dhcp4 -c /etc/kea/kea-dhcp4.conf (code=exited, status=0/SUCCESS)
   Main PID: 870 (code=exited, status=0/SUCCESS)
   Mem peak: 11.2M
   CPU: 868ms

d'oct. 05 21:00:09 kea-server-shailmartinez systemd[1]: Started kea-dhcp4-server.service - Kea IPv4 DHCP daemon.
d'oct. 05 21:00:13 kea-server-shailmartinez (kea-dhcp4)[870]: kea-dhcp4-server.service: ConfigurationDirectory 'kea' already exists but the mode is differ
d'oct. 05 21:00:15 kea-server-shailmartinez kea-dhcp4[870]: 2026-10-05 21:00:15.319 INFO [kea-dhcp4.dhcp4/870.138479315643520] DHCP4_STARTING Kea DHCPv4
d'oct. 05 21:00:15 kea-server-shailmartinez kea-dhcp4[870]: 2026-10-05 21:00:15.414 INFO [kea-dhcp4.commands/870.138479315643520] COMMAND_RECEIVED Receive
d'oct. 05 21:25:14 kea-server-shailmartinez systemd[1]: Stopping kea-dhcp4-server.service - Kea IPv4 DHCP daemon...
d'oct. 05 21:25:14 kea-server-shailmartinez systemd[1]: kea-dhcp4-server.service: Deactivated successfully.
d'oct. 05 21:25:14 kea-server-shailmartinez systemd[1]: Stopped kea-dhcp4-server.service - Kea IPv4 DHCP daemon.

smartinez6@kea-server-shailmartinez:~$ sudo systemctl status kea-ctrl-agent
o kea-ctrl-agent.service - Kea Control Agent
   Loaded: loaded (/usr/lib/systemd/system/kea-ctrl-agent.service; enabled; preset: enabled)
   Active: inactive (dead) since Mon 2026-10-05 21:25:19 UTC; 1min 54s ago
   Duration: 25min 9.504s
   Invocation: 13b61595a7a4365b73e6db36611a9cd
   Docs: man:kea-ctrl-agent(8)
   Process: 869 ExecStart=/usr/sbin/kea-ctrl-agent -c /etc/kea/kea-ctrl-agent.conf (code=exited, status=0/SUCCESS)
   Main PID: 869 (code=exited, status=0/SUCCESS)
   Mem peak: 6.2M
   CPU: 217ms

d'oct. 05 21:00:09 kea-server-shailmartinez systemd[1]: Started kea-ctrl-agent.service - Kea Control Agent.
d'oct. 05 21:00:13 kea-server-shailmartinez (kea-ctrl-agent)[869]: kea-ctrl-agent.service: ConfigurationDirectory 'kea' already exists but the mode is dif
d'oct. 05 21:00:15 kea-server-shailmartinez kea-ctrl-agent[869]: 2026-10-05 21:00:15.187 INFO [kea-ctrl-agent.dctl/869.123290438102080] DCTL_STARTING Comp
d'oct. 05 21:25:19 kea-server-shailmartinez systemd[1]: Stopping kea-ctrl-agent.service - Kea Control Agent...
d'oct. 05 21:25:19 kea-server-shailmartinez systemd[1]: kea-ctrl-agent.service: Deactivated successfully.
d'oct. 05 21:25:19 kea-server-shailmartinez systemd[1]: Stopped kea-ctrl-agent.service - Kea Control Agent.
[lines 1-17/17 (END)]
```

Aquest és el clon (server secundari) amb IP 172.24.63.11

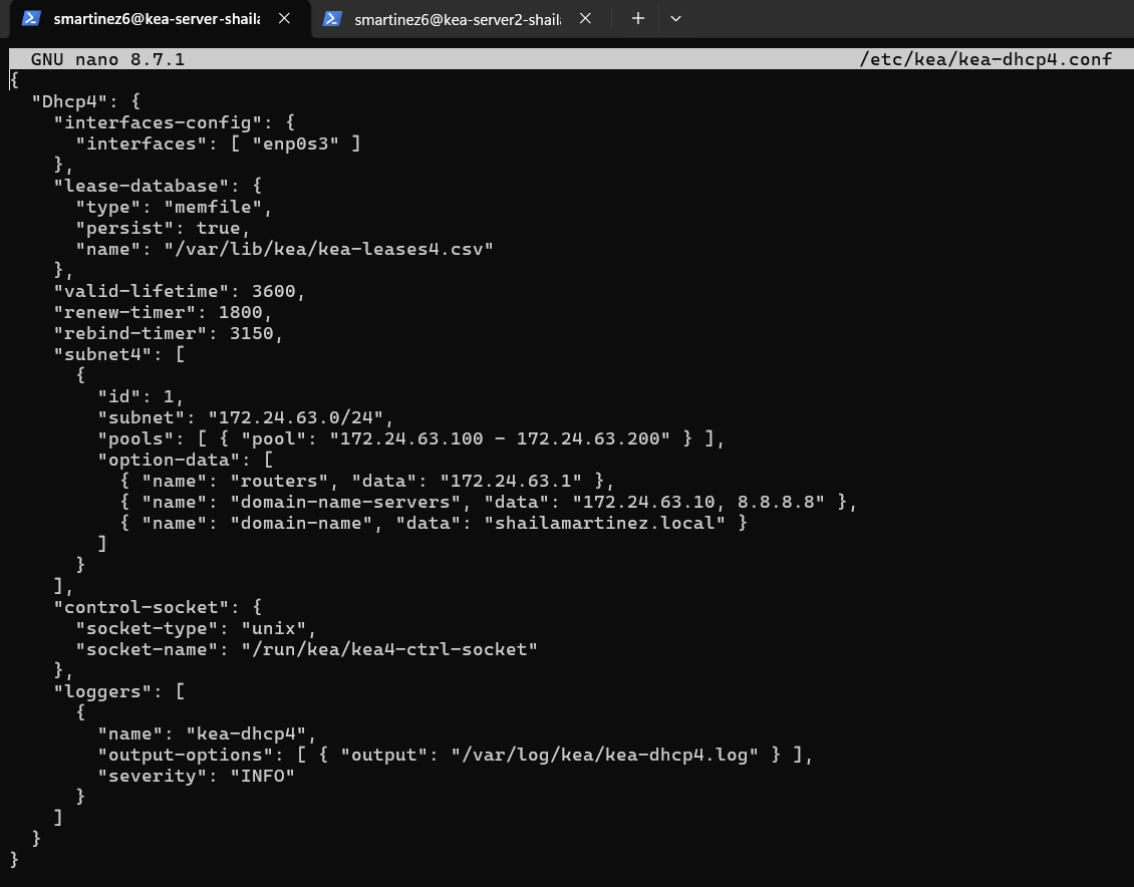
```
smartinez6@kea-server-shaila x smartinez6@kea-server2-shai x + v
smartinez6@kea-server2-shailamartinez:~$ hostnamectl
Static hostname: kea-server2-shailamartinez
Icon name: computer-vm
Chassis: vm
Machine ID: 46dd04e208994a328d9dc0fc92eba10b
Boot ID: 0db650fd3f5640bb9a66e305a12befb0
Virtualization: oracle
Operating System: Ubuntu 26.04.1 LTS
Kernel: Linux 7.0.0-34-generic
Architecture: x86_64
Hardware Vendor: innotek GmbH
Hardware Model: VirtualBox
Hardware Version: 1.2
Firmware Version: VirtualBox
Firmware Date: Fri 2006-12-01
Firmware Age: 19y 10month 4d
smartinez6@kea-server2-shailamartinez:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:0e:86:06 brd ff:ff:ff:ff:ff:ff
    altname enx0800270e8606
    inet 172.24.63.11/24 brd 172.24.63.255 scope global enp0s3
        valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fe0e:8606/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
```

Servei de Kea inactiu.

```
smartinez6@kea-server-shaila x smartinez6@kea-server2-shai x + v
smartinez6@kea-server2-shailamartinez:~$ sudo systemctl stop kea-dhcp4-server
smartinez6@kea-server2-shailamartinez:~$ sudo systemctl stop kea-ctrl-agent
smartinez6@kea-server2-shailamartinez:~$ sudo systemctl status kea-dhcp4-server
o kea-dhcp4-server.service - Kea IPv4 DHCP daemon
   Loaded: loaded (/usr/lib/systemd/system/kea-dhcp4-server.service; enabled; preset: enabled)
   Active: inactive (dead) since Mon 2026-10-05 21:28:08 UTC; 25s ago
   Duration: 24min 52.580s
   Invocation: 9450d0ff0181490188874daa3fd01039
   Docs: man:kea-dhcp4(8)
   Process: 1149 ExecStart=/usr/sbin/kea-dhcp4 -c /etc/kea/kea-dhcp4.conf (code=exited, status=0/SUCCESS)
   Main PID: 1149 (code=exited, status=0/SUCCESS)
   Mem peak: 10.1M
   CPU: 316ms
smartinez6@kea-server2-shailamartinez:~$ sudo systemctl status kea-ctrl-agent
o kea-ctrl-agent.service - Kea Control Agent
   Loaded: loaded (/usr/lib/systemd/system/kea-ctrl-agent.service; enabled; preset: enabled)
   Active: inactive (dead) since Mon 2026-10-05 21:28:27 UTC; 22s ago
   Duration: 25min 11.167s
   Invocation: 2aa9527a0a4b49fd9b1a340f8e7ddec
   Docs: man:kea-ctrl-agent(8)
   Process: 1148 ExecStart=/usr/sbin/kea-ctrl-agent -c /etc/kea/kea-ctrl-agent.conf (code=exited, status=0/SUCCESS)
   Main PID: 1148 (code=exited, status=0/SUCCESS)
   Mem peak: 7.2M
   CPU: 96ms
smartinez6@kea-server2-shailamartinez:~$ |
```

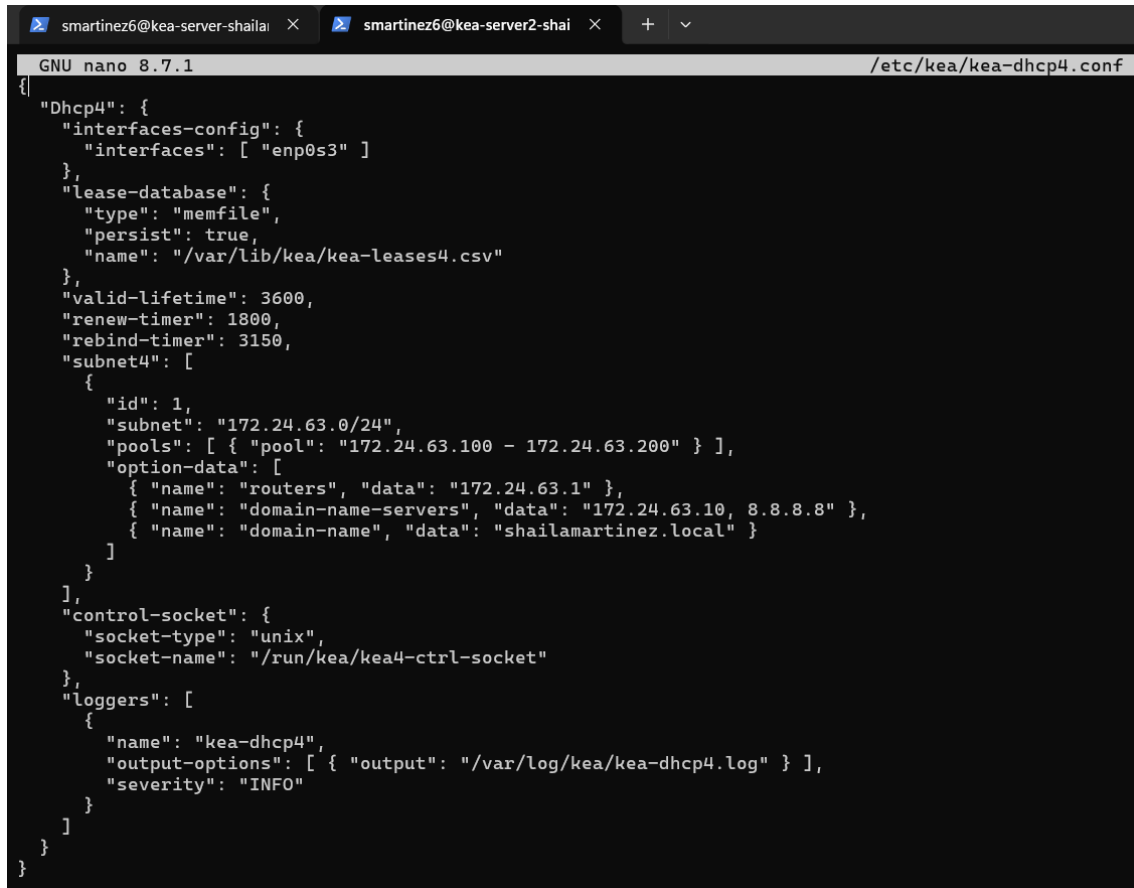
Part 2: Configuració Base Idèntica del Subnet

Servidor primari amb la configuració del kea-dhcp4



```
GNU nano 8.7.1 /etc/kea/kea-dhcp4.conf
{
  "Dhcp4": {
    "interfaces-config": {
      "interfaces": [ "enp0s3" ]
    },
    "lease-database": {
      "type": "memfile",
      "persist": true,
      "name": "/var/lib/kea/kea-leases4.csv"
    },
    "valid-lifetime": 3600,
    "renew-timer": 1800,
    "rebind-timer": 3150,
    "subnet4": [
      {
        "id": 1,
        "subnet": "172.24.63.0/24",
        "pools": [ { "pool": "172.24.63.100 - 172.24.63.200" } ],
        "option-data": [
          { "name": "routers", "data": "172.24.63.1" },
          { "name": "domain-name-servers", "data": "172.24.63.10, 8.8.8.8" },
          { "name": "domain-name", "data": "shailamartinez.local" }
        ]
      }
    ],
    "control-socket": {
      "socket-type": "unix",
      "socket-name": "/run/kea/kea4-ctrl-socket"
    },
    "loggers": [
      {
        "name": "kea-dhcp4",
        "output-options": [ { "output": "/var/log/kea/kea-dhcp4.log" } ],
        "severity": "INFO"
      }
    ]
  }
}
```

Servidor secundari amb la mateixa IP, i el contingut exacte que el primer perquè pugui sincronitzar les concessions.



```
GNU nano 8.7.1 /etc/kea/kea-dhcp4.conf
{
  "Dhcp4": {
    "interfaces-config": {
      "interfaces": [ "enp0s3" ]
    },
    "lease-database": {
      "type": "memfile",
      "persist": true,
      "name": "/var/lib/kea/kea-leases4.csv"
    },
    "valid-lifetime": 3600,
    "renew-timer": 1800,
    "rebind-timer": 3150,
    "subnet4": [
      {
        "id": 1,
        "subnet": "172.24.63.0/24",
        "pools": [ { "pool": "172.24.63.100 - 172.24.63.200" } ],
        "option-data": [
          { "name": "routers", "data": "172.24.63.1" },
          { "name": "domain-name-servers", "data": "172.24.63.10, 8.8.8.8" },
          { "name": "domain-name", "data": "shailamartinez.local" }
        ]
      }
    ]
  },
  "control-socket": {
    "socket-type": "unix",
    "socket-name": "/run/kea/kea4-ctrl-socket"
  },
  "loggers": [
    {
      "name": "kea-dhcp4",
      "output-options": [ { "output": "/var/log/kea/kea-dhcp4.log" } ],
      "severity": "INFO"
    }
  ]
}
```

Mateixa configuració de kea-ctrl-agent als dos servers. Escolta a 0.0.0.0:8000



```
GNU nano 8.7.1 /etc/kea/kea-ctrl-agent.conf
{
  "Control-agent": {
    "http-host": "0.0.0.0",
    "http-port": 8000,
    "control-sockets": {
      "dhcp4": {
        "socket-type": "unix",
        "socket-name": "/run/kea/kea4-ctrl-socket"
      }
    }
  }
}
```

Reinici i estat dels serveis als dos servidors.

```
smartinez6@kea-server-shaila x smartinez6@kea-server2-shaila x + v
smartinez6@kea-server-shailamartinez:~$ sudo systemctl restart kea-dhcp4-server
smartinez6@kea-server-shailamartinez:~$ sudo systemctl restart kea-ctrl-agent
smartinez6@kea-server-shailamartinez:~$ sudo systemctl status kea-ctrl-agent
● kea-ctrl-agent.service - Kea Control Agent
   Loaded: loaded (/usr/lib/systemd/system/kea-ctrl-agent.service; enabled; preset: enabled)
   Active: active (running) since Mon 2026-10-05 22:38:31 UTC; 10s ago
 Invocation: b79b3552729f41dea42c22b7742c43a
    Docs: man:kea-ctrl-agent(8)
   Main PID: 3911 (kea-ctrl-agent)
     Tasks: 5 (limit: 1891)
    Memory: 1.6M (peak: 2.1M)
       CPU: 122ms
   CGroup: /system.slice/kea-ctrl-agent.service
           └─3911 /usr/sbin/kea-ctrl-agent -c /etc/kea/kea-ctrl-agent.conf

d'oct. 05 22:38:31 kea-server-shailamartinez systemd[1]: Started kea-ctrl-agent.service - Kea Control Agent.
d'oct. 05 22:38:31 kea-server-shailamartinez kea-ctrl-agent[3911]: 2026-10-05 22:38:31.802 INFO [kea-ctrl-agent.dctl/3>
d'oct. 05 22:38:31 kea-server-shailamartinez kea-ctrl-agent[3911]: 2026-10-05 22:38:31.803 INFO [kea-ctrl-agent.ctrl-a>
d'oct. 05 22:38:31 kea-server-shailamartinez kea-ctrl-agent[3911]: 2026-10-05 22:38:31.803 INFO [kea-ctrl-agent.dctl/3>
d'oct. 05 22:38:31 kea-server-shailamartinez kea-ctrl-agent[3911]: 2026-10-05 22:38:31.806 INFO [kea-ctrl-agent.ctrl-a>
d'oct. 05 22:38:31 kea-server-shailamartinez kea-ctrl-agent[3911]: 2026-10-05 22:38:31.807 WARN [kea-ctrl-agent.ctrl-a>

smartinez6@kea-server-shailamartinez:~$ sudo systemctl status kea-dhcp4-server
● kea-dhcp4-server.service - Kea IPv4 DHCP daemon
   Loaded: loaded (/usr/lib/systemd/system/kea-dhcp4-server.service; enabled; preset: enabled)
   Active: active (running) since Mon 2026-10-05 22:38:23 UTC; 30s ago
 Invocation: 9f47ab94c6ad41dca56812dcb5afea50
    Docs: man:kea-dhcp4(8)
   Main PID: 3885 (kea-dhcp4)
     Tasks: 7 (limit: 1891)
    Memory: 2.3M (peak: 2.5M)
       CPU: 171ms
   CGroup: /system.slice/kea-dhcp4-server.service
           └─3885 /usr/sbin/kea-dhcp4 -c /etc/kea/kea-dhcp4.conf

d'oct. 05 22:38:23 kea-server-shailamartinez systemd[1]: Started kea-dhcp4-server.service - Kea IPv4 DHCP daemon.
d'oct. 05 22:38:24 kea-server-shailamartinez kea-dhcp4[3885]: 2026-10-05 22:38:24.087 INFO [kea-dhcp4.dhcp4/3885.14040>
d'oct. 05 22:38:24 kea-server-shailamartinez kea-dhcp4[3885]: 2026-10-05 22:38:24.089 INFO [kea-dhcp4.commands/3885.14>

smartinez6@kea-server-shailamartinez:~$ |
```

Els dos apareixen en active running-

```
smartinez6@kea-server-shaila x smartinez6@kea-server2-shai x + v
smartinez6@kea-server2-shailamartinez:~$ sudo systemctl status kea-dhcp4-server
● kea-dhcp4-server.service - Kea IPv4 DHCP daemon
   Loaded: loaded (/usr/lib/systemd/system/kea-dhcp4-server.service; enabled; preset: enabled)
   Active: active (running) since Mon 2026-10-05 22:45:05 UTC; 12s ago
 Invocation: c0ae28e742c94d4581329c6ee5c242e2
    Docs: man:kea-dhcp4(8)
   Main PID: 4269 (kea-dhcp4)
     Tasks: 7 (limit: 1891)
    Memory: 2.6M (peak: 2.7M)
       CPU: 139ms
   CGroup: /system.slice/kea-dhcp4-server.service
           └─4269 /usr/sbin/kea-dhcp4 -c /etc/kea/kea-dhcp4.conf

smartinez6@kea-server2-shailamartinez:~$ sudo systemctl status kea-ctrl-agent
● kea-ctrl-agent.service - Kea Control Agent
   Loaded: loaded (/usr/lib/systemd/system/kea-ctrl-agent.service; enabled; preset: enabled)
   Active: active (running) since Mon 2026-10-05 22:44:13 UTC; 1min 15s ago
 Invocation: 1f6606d2088949ee836a0082be2347be
    Docs: man:kea-ctrl-agent(8)
   Main PID: 4230 (kea-ctrl-agent)
     Tasks: 5 (limit: 1891)
    Memory: 1.6M (peak: 2.2M)
       CPU: 93ms
   CGroup: /system.slice/kea-ctrl-agent.service
           └─4230 /usr/sbin/kea-ctrl-agent -c /etc/kea/kea-ctrl-agent.conf

smartinez6@kea-server2-shailamartinez:~$ |
```

Com el HA només sincronitza les concessions i no la configuració, si els dos servers donessin opcions diferents, el client podria perdre la connexió després del failover i no funcionaria correctament.

Server 2: És el mateix bloc primari però “this-server-name” canviat pel nom “server2” i els peers idèntics.

Comprovació status-get en rol stand-by i l'estat syncing i el partnet "server1". El secundari ja reconeix el primari i està sincronitzant les concessions.

```

smartinez@kea-server-shaila x smartinez@kea-server-shal x
smartinez@kea-server-shaila:~$ sudo kea-dhcpd -t /etc/kea/kea-dhcpd.conf
2026-10-06 09:18:24.089 INFO [kea-dhcpd/3165.139548285922432] HOOKS_LIBRARY_CLOSED hooks library /usr/lib/x86_64-linux-gnu/kea/hooks/libdhcp_ha.so successfully closed
2026-10-06 09:18:24.089 INFO [kea-dhcpd.dhcpdsvr/3165.139548285922432] DHCPDRV_CFGMGR_NEW_SUBNET4 a new subnet has been added to configuration: 172.24.63.0/24 with params
t1=1800, t2=3150, valid-lifetime=3600
2026-10-06 09:18:24.089 INFO [kea-dhcpd.dhcpdsvr/3165.139548285922432] DHCPDRV_CFGMGR_SOCKET_TYPE_SELECT using socket type raw
2026-10-06 09:18:24.090 INFO [kea-dhcpd.dhcpdsvr/3165.139548285922432] DHCPDRV_CFGMGR_ADD_IFACE listening on interface enp0s3
2026-10-06 09:18:24.090 INFO [kea-dhcpd.dhcpdsvr/3165.139548285922432] DHCPDRV_CFGMGR_SOCKET_TYPE_DEFAULT "dhcp-socket-type" not specified, using default socket type raw
2026-10-06 09:18:24.090 INFO [kea-dhcpd.dhcpdsvr/3165.139548285922432] DHCPDRV_LEASE_MGR_BACKENDS_REGISTERED the following lease backend types are available: memfile
2026-10-06 09:18:24.090 INFO [kea-dhcpd.hosts/3165.139548285922432] HOSTS_BACKENDS_REGISTERED the following host backend types are available:
2026-10-06 09:18:24.090 INFO [kea-dhcpd.dhcpdsvr/3165.139548285922432] DHCPDRV_FORENSIC_BACKENDS_REGISTERED the following forensic backend types are available:
2026-10-06 09:18:24.090 INFO [kea-dhcpd.database/3165.139548285922432] CONFIG_BACKENDS_REGISTERED the following config backend types are available:
smartinez@kea-server-shaila:~$ sudo systemctl restart kea-dhcpd-server
smartinez@kea-server-shaila:~$ sudo systemctl status kea-dhcpd-server
● kea-dhcpd-server.service - Kea IPv4 DHCP daemon
   Loaded: loaded (/usr/lib/systemd/system/kea-dhcpd-server.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-10-06 09:18:40 UTC; 6s ago
   Invocation: c3b3a162ff2413da15035a5fabbb086
   Docs: man:kea-dhcpd(8)
   Main PID: 3165 (kea-dhcpd)
   Tasks: 5 (limit: 1091)
   Memory: 2.6M (peak: 2.6M)
   CPU: 31ms
   CGroup: /system.slice/kea-dhcpd-server.service
           └─3165 /usr/sbin/kea-dhcpd -c /etc/kea/kea-dhcpd.conf

d'oct. 06 09:18:40 kea-server-shailamartinez systemd[1]: Started Kea dhcpd-server service - Kea IPv4 DHCP daemon.
d'oct. 06 09:18:40 kea-server-shailamartinez kea-dhcpd[3165]: 2026-10-06 09:18:40.600 INFO [kea-dhcpd.configmgr/3165.139548285922432]
d'oct. 06 09:18:40 kea-server-shailamartinez kea-dhcpd[3165]: 2026-10-06 09:18:40.603 INFO [kea-dhcpd.configmgr/3165.139548285922432]

```

```
smartinez6@kea-server-shaila:~$ curl -X POST -H "Content-Type: application/json" \
-d '{"command": "status-get", "service": ["dhcp4"]}' \
http://172.24.63.11:8000/
[{"arguments": {"dhcp-state": {"disabled-by-db-connection": [ ], "disabled-by-local-command": [ ], "disabled-by-remote-command": [ ], "disabled-by-user": false, "globally-disabled": false }, "high-availability": { "ha-mode": "hot-standby", "ha-servers": { "local": { "role": "standby", "scopes": [ ], "server-name": "server2", "state": "hot-standby", "system-time": "2026-10-07 16:33:13", "remote": { "age": 8, "analyzed-packets": 0, "clock-skew": 0, "communication-interrupted": false, "connecting-clients": 0, "in-touch": true, "last-scopes": [ "server1" ], "last-state": "hot-standby", "role": "primary", "server-name": "server1", "system-time": "2026-10-07 16:33:13", "unacked-clients": 0, "unacked-clients-left": 0 } } }, "multi-threading-enabled": false, "pid": 934, "reload": 644, "sockets": { "status": "ready", "uptime": 645 }, "result": 0 } ]
smartinez6@kea-server2-shailamartinez:~$
```

Part 4: Verificació del Failover i Failback

Failover (caiguda del primari)

Sol·licitud de concessió normal al client i mostra la IP del pool (servidor primari).
Aturo el servidor i es mostra inactiu.

```
smartinez6@kea-server-shaila:~$ sudo systemctl stop kea-dhcp4-server
[sudo: authenticate] Password:
smartinez6@kea-server-shailamartinez:~$ sudo systemctl status kea-dhcp4-server
o kea-dhcp4-server.service - Kea IPv4 DHCP daemon
   Loaded: loaded (/usr/lib/systemd/system/kea-dhcp4-server.service; enabled; preset: enabled)
   Active: inactive (dead) since Tue 2026-10-06 09:58:57 UTC; 56s ago
   Duration: 49min 299ms
   Invocation: 776c5cf81c884a1194d7b5f5834ded86e
   Docs: man:kea-dhcp4(8)
   Process: 3900 ExecStart=/usr/sbin/kea-dhcp4 -c /etc/kea/kea-dhcp4.conf (code=exited, status=0/SUCCESS)
   Main PID: 3900 (code=exited, status=0/SUCCESS)
   Mem peak: 3.1M
   CPU: 1.152s

d'oct. 06 09:09:57 kea-server-shailamartinez systemd[1]: Started kea-dhcp4-server.service - Kea IPv4 DHCP daemon.
d'oct. 06 09:09:57 kea-server-shailamartinez kea-dhcp4[3900]: 2026-10-06 09:09:57.079 INFO [kea-dhcp4.dhcp4/3900.12969>
d'oct. 06 09:09:57 kea-server-shailamartinez kea-dhcp4[3900]: 2026-10-06 09:09:57.080 INFO [kea-dhcp4.commands/3900.12>
d'oct. 06 09:58:57 kea-server-shailamartinez systemd[1]: Stopping kea-dhcp4-server.service - Kea IPv4 DHCP daemon...
d'oct. 06 09:58:57 kea-server-shailamartinez systemd[1]: kea-dhcp4-server.service: Deactivated successfully.
d'oct. 06 09:58:57 kea-server-shailamartinez systemd[1]: Stopped kea-dhcp4-server.service - Kea IPv4 DHCP daemon.
d'oct. 06 09:58:57 kea-server-shailamartinez systemd[1]: kea-dhcp4-server.service: Consumed 1.152s CPU time over 49min >

smartinez6@kea-server-shailamartinez:~$
```

```
SMG-UCT [S'està executant] - Oracle VirtualBox
Fitxer Màquina Visualitza Entrada Dispositius Ajuda

6 de oct 12:00
smartinez6@SMG-UD1:~
smartinez6@SMG-UD1:~$ sudo dhclient -r enp0s3
[sudo: authenticate] Contraseña:
smartinez6@SMG-UD1:~$ sudo dhclient enp0s3
Setting LLNMR support level "yes" for "2", but the global support level is "no".
smartinez6@SMG-UD1:~$ ip a show enp0s3
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:11:22:33 brd ff:ff:ff:ff:ff:ff
    altname enx080027112233
    inet 172.24.63.100/24 brd 172.24.63.255 scope global dynamic enp0s3
        valid lft 3304sec preferred lft 3344sec
    inet6 fe80::a00:27ff:fe11:2233/64 scope link noprefixroute
        valid lft forever preferred lft forever
smartinez6@SMG-UD1:~$
```

Status-get del secundari, ha passat a partnet-down. Ha detectat que el primari no respon i assumeix el servei del pool. El client renova la IP obtenint la adreça del pool i no perd la connectivitat.

```
smartinez6@kea-server-shaila:~$ curl -X POST -H "Content-Type: application/json" \
-d '{"command": "status-get", "service": ["dhcp4"]}' \
http://172.24.63.11:8000/
[{"arguments": {"dhcp-state": {"disabled-by-db-connection": [ ], "disabled-by-local-command": [ ], "disabled-by-remote-command": [ ], "disabled-by-user": false, "globally-disabled": false }, "high-availability": { "ha-mode": "hot-standby", "ha-servers": { "local": { "role": "standby", "scopes": [ "serv", "server1" ], "server-name": "server2", "state": "partner-down", "system-time": null }, "remote": { "age": 224, "analyzed-packets": 0, "clock-skew": null, "communication-interrupted": true, "connecting-clients": 0, "in-touch": true, "last-scopes": [ "server1" ], "last-state": "unavailable", "role": "primary", "server-name": "server1", "system-time": null, "unacked-clients": 0, "unacked-clients-left": 1 } } }, "multi-threading-enabled": false, "pid": 3185, "reload": 26
smartinez6@kea-server2-shailamartinez:~$
```

```
SMG-UCT [S'està executant] - Oracle VirtualBox
Fitxer Màquina Visualitza Entrada Dispositius Ajuda

6 de oct 12:04
smartinez6@SMG-UD1:~
smartinez6@SMG-UD1:~$ sudo dhclient -r enp0s3
killed old client process
smartinez6@SMG-UD1:~$ sudo dhclient enp0s3
Setting LLNMR support level "yes" for "2", but the global support level is "no".
smartinez6@SMG-UD1:~$ ip a show enp0s3
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:11:22:33 brd ff:ff:ff:ff:ff:ff
    altname enx080027112233
    inet 172.24.63.100/24 brd 172.24.63.255 scope global dynamic enp0s3
        valid lft 3004sec preferred lft 3004sec
    inet6 fe80::a00:27ff:fe11:2233/64 scope link noprefixroute
        valid lft forever preferred lft forever
smartinez6@SMG-UD1:~$
```


Failback (recuperació del primari)

Primari: inicio el servei i mostra hot-standby role standby, i remot del serveren hot-standby i el remot del server2 també. El primari ha recuperat el seu rol.

```
smartinez6@kea-server-shailamartinez:~$ sudo systemctl start kea-dhcp4-server
smartinez6@kea-server-shailamartinez:~$ sudo systemctl status kea-dhcp4-server
● kea-dhcp4-server.service - Kea IPv4 DHCP daemon
   Loaded: loaded (/usr/lib/systemd/system/kea-dhcp4-server.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-10-06 10:44:51 UTC; 3min 18s ago
  Invocation: e334833fdb4140b597ef5a2e7525f992
     Docs: man:kea-dhcp4(8)
    Main PID: 4490 (kea-dhcp4)
      Tasks: 5 (limit: 1891)
     Memory: 2.5M (peak: 3M)
        CPU: 81ms
    CGroup: /system.slice/kea-dhcp4-server.service
            └─4490 /usr/sbin/kea-dhcp4 -c /etc/kea/kea-dhcp4.conf

d'oct. 06 10:44:51 kea-server-shailamartinez systemd[1]: Started kea-dhcp4-server.service - Kea IPv4 DHCP daemon.
d'oct. 06 10:44:51 kea-server-shailamartinez kea-dhcp4[4490]: 2026-10-06 10:44:51.293 INFO [kea-dhcp4.dhcp4/4490.132755]
d'oct. 06 10:44:51 kea-server-shailamartinez kea-dhcp4[4490]: 2026-10-06 10:44:51.294 INFO [kea-dhcp4.commands/4490.132755]

smartinez6@kea-server-shailamartinez:~$ curl -s -X POST -H "Content-Type: application/json" \
-d '{"command": "status-get", "service": ["dhcp4"]}' \
http://172.24.63.10:8000/
[{"arguments": {"dhcp-state": {"disabled-by-db-connection": [ ], "disabled-by-local-command": [ ], "disabled-by-remote-command": [ ], "disabled-by-user": false, "globally-disabled": false }, "high-availability": [ { "ha-mode": "hot-standby", "ha-servers": { "local": { "role": "primary", "scopes": [ "server1" ], "server-name": "server1", "state": "hot-standby", "system-time": "2026-10-06 10:48:23" }, "remote": { "age": 0, "analyzed-packets": 0, "clock-skew": 0, "communication-interrupted": false, "connecting-clients": 0, "in-touch": true, "last-scopes": [ ], "last-state": "hot-standby", "role": "standby", "server-name": "server2", "system-time": "2026-10-06 10:48:23", "unacked-clients": 0, "unacked-clien
```

Secundari: state: hot-standby, role: standby i el remot server1 en hot-standby. Els dos servers tornen a l'estat normal.

```
smartinez6@kea-server2-shailamartinez:~$ sudo systemctl status kea-dhcp4-server.service
● kea-dhcp4-server.service - Kea IPv4 DHCP daemon
   Loaded: loaded (/usr/lib/systemd/system/kea-dhcp4-server.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-10-06 10:44:35 UTC; 5min ago
  Invocation: 528a9de4871f49319138dcecf6c91159
     Docs: man:kea-dhcp4(8)
    Main PID: 3567 (kea-dhcp4)
      Tasks: 5 (limit: 1891)
     Memory: 2.8M (peak: 3M)
        CPU: 99ms
    CGroup: /system.slice/kea-dhcp4-server.service
            └─3567 /usr/sbin/kea-dhcp4 -c /etc/kea/kea-dhcp4.conf

d'oct. 06 10:44:35 kea-server2-shailamartinez systemd[1]: Started kea-dhcp4-server.service - Kea IPv4 DHCP daemon.
d'oct. 06 10:44:36 kea-server2-shailamartinez kea-dhcp4[3567]: 2026-10-06 10:44:36.033 INFO [kea-dhcp4.dhcp4/3567.13085]
d'oct. 06 10:44:36 kea-server2-shailamartinez kea-dhcp4[3567]: 2026-10-06 10:44:36.034 INFO [kea-dhcp4.commands/3567.13085]

smartinez6@kea-server2-shailamartinez:~$ curl -s -X POST -H "Content-Type: application/json" \
-d '{"command": "status-get", "service": ["dhcp4"]}' \
http://172.24.63.11:8000/
[{"arguments": {"dhcp-state": {"disabled-by-db-connection": [ ], "disabled-by-local-command": [ ], "disabled-by-remote-command": [ ], "disabled-by-user": false, "globally-disabled": false }, "high-availability": [ { "ha-mode": "hot-standby", "ha-servers": { "local": { "role": "standby", "scopes": [ ], "server-name": "server2", "state": "hot-standby", "system-time": "2026-10-06 10:49:34" }, "remote": { "age": 10, "analyzed-packets": 0, "clock-skew": 0, "communication-interrupted": false, "connecting-clients": 0, "in-touch": true, "last-scopes": [ "server1" ], "last-state": "hot-standby", "role": "primary", "server-name": "server1", "system-time": "2026-10-06 10:49:34", "unacked-clients": 0, "unacked-clients-left": 0 } } }, {"multi-threading-enabled": false, "pid": 3567, "reload": 309, "sockets": { "status": "ready", "uptime": 309 }, "result": 0 } ]smartinez6@kea-server2-shailamartinez:~$
```

Qüestions

1. Quina diferència hi ha entre els modes hot-standby i load-balancing del hook libdhcp_ha.so? En quin escenari triaríeu cadascun (per exemple, si el servidor secundari té menys capacitat que el primari)?

El hot-standby respon al primari, el secundari espera i només actua si el primari cau. El load-balance responen els dos i es reparteixen els clients. Escolliria hot-standby en el cas de que el secundari té menys capacitat. Sino load balancing si els dos són similars.

2. Per què cal que la configuració de subnet4 (rangs, opcions) sigui idèntica als dos servidors, mentre que el bloc high-availability és l'únic que canvia entre ells?

El HA només sincronitza les concessions i no la configuració, si fossin diferents la màquina client rebria opcions diferents segons qui respon. El HA només canvia "this-server-name" que és el que indica qui es cada servidor.

3. Quins estats pot mostrar un servidor Kea en una relació HA (waiting, syncing, ready, partner-down, etc.) i què indica cadascun sobre la salut de la parella?

Waiting: acaba d'arrencar i espera el segon per veure en quin estat està.

Syncing: sincronitzant les concessions amb el segon

Ready: ha acabat de sincronitzar i està preparat

Partner-down: el segon no respon i el dona per caigut, i aquest servidor atén sol a tots els clients.

Hot-standby load-balancing: estat normal del mode configurat i els dos servidors es veuen.

Communication-recovery: és quan s'ha perdut la comunicació però el segon podria estar en bon estat. Els dos funcionen mentres intenten recuperarla (només ne load balancing).

communication-interrupted: la comunicació amb el segon esta interrompuda però encara no està caigut.

Terminated: el servidor deixa de funcionar en HA i és un error crític (desincronització del rellotge entre els dos servers).

In-maintenance: servidor en manteniment a propòsit, el segon assumeix el servei i no el considera una caiguda.

4. Compareu aquest mecanisme amb el failover peer clàssic d'ISC DHCP (vist a l'activitat d'aprofundiment AP0208): quins avantatges aporta el hook natiu de Kea en termes d'observabilitat i facilitat de configuració?

És més fàcil de configurar Kea i més observable (json peers igual als dos servers), ja que status-get permet veure l'estat en temps real mentre que en ISC s'ha de mirar els logs i està desfasat.

He tingut diversos problemes amb conflictes amb la comunicació HTTP pel port 8000 entre els servidors i he afegit una llibreria extra al kea-dhcp4.conf i algunes opcions més, totes explicades al README.md